

Geometry & Lattices

Preface

These are a collection of notes from *Sphere Packings, Lattices and Groups* (by J.H. Conway and N.J.A. Sloane) and various other resources. I would like to thank Professor Richard E. Borcherds for his helpful comments and suggestions.

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1 Introduction

A *lattice packing* is a packing of spheres so that 0 is a center and that if there are spheres with centers u and v , then there are also spheres with centers at $u + v$ and $u - v$. Thus, we see that the centers of the spheres form an additive group, which can be visualized easily in the dimension 2 case. Take now an n -dimensional lattice with some centers $v_1, \dots, v_n$ so that the set of all centers can be constructing by enumerating $\sum k_i v_i$, with $k_i \in \mathbb{Z}$. In other words, suppose the list $v_1, \dots, v_n$ spans the the set of all centers of the lattice. Then we call the $v_1, \dots, v_n$ a basis for the lattice.

As a vague side note, we use the term *parallelotope* (in dimension > 3) as a generalization of parallelepiped in dimension 3. This is similar to how parallelepiped is a generalization of parallelogram.

Suppose now $v_1, \dots, v_n$ is a basis for a lattice. The parallelotope consisting of the points $\theta_1 v_1 + \dots + \theta_n v_n$ ($0 \leq \theta_i < 1$) is called a fundamental region. We now consider a trivial example. In $\mathbb{R}^2$, denote Λ by the lattice of points with centers precisely the set $\mathbb{Z}^2$. Then, a 1×1 square with lattice points as its vertices is a fundamental region.

We now consider an important quantity. By the term *determinant* or *discriminant*, we refer to the square of the volume of the fundamental region of a lattice Λ . We may calculate the determinant of a lattice. Suppose Λ is a lattice with basis $v_1, \dots, v_n$. Write the basis vectors as follows:

$$v_1 = (v_{11}, v_{12}, \dots, v_{1m})$$

$$v_2 = (v_{21}, v_{22}, \dots, v_{2m})$$

$$\dots$$

$$v_n = (v_{n1}, v_{n2}, \dots, v_{nm})$$

where $m \geq n$. (Note that m is the dimension of the space, whereas n is the dimension of the lattice.) Put a matrix M with rows the basis vectors:

$$M = \begin{pmatrix} v_{11} & v_{12} & \cdots & v_{1m} \\ v_{21} & v_{22} & \cdots & v_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ v_{n1} & v_{n2} & \cdots & v_{nm} \end{pmatrix}$$

M is called the *generator matrix* for the lattice. Note also that the lattice vectors consist of all vectors ζM , where $\zeta = (\zeta_1, \dots, \zeta_n)$ is an arbitrary vector with $\zeta_i \in \mathbb{Z}$ for all i . The matrix $A = MM^t$ is called the *Gram matrix* for the lattice. (As a side note, inner products will be assumed to refer to the standard dot product.)

Theorem. The determinant of a lattice Λ is the determinant of A .

Note also that if M is square, then $\det M = \det M^t$, so the equation $\det \Lambda = \det A$ gives $\det \Lambda = \det A = \det(MM^t) = \det M \cdot \det M^t = (\det M)^2$. We summarize this in the following proposition.

Proposition. If M is a square matrix, then $\det \Lambda = (\det M)^2$.

We now consider an example. Put Λ as the planar hexagonal lattice in $\mathbb{R}^2$ and we wish to compute $\det \Lambda$. Choosing a convenient point in the lattice to be 0, we have that a basis is $v_1 = (1, 0), v_2 = (1/2, \sqrt{3}/2)$. Thus, $M := \begin{pmatrix} 1 & 0 \\ 1/2 & \sqrt{3}/2 \end{pmatrix}$. M is square, so $\det \Lambda = (\det M)^2 = 3/4$.

Note that we could also assigned to M the matrix

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$$

Lattices Λ_1, Λ_2 with generator matrices M_1, M_2 (respectively) are *equivalent* if they describe the same lattice but in different coordinates and scale. We will shortly include a more precise definition for lattice equivalence.

The *density* Δ of a lattice Λ is the proportion of the space that is occupied by the spheres, which is equal to the volume of one sphere divided by the volume of the fundamental region, which is equal to the volume of one sphere divided by the square root of $\det \Lambda$.

Consider now the hexagonal packing of spheres (in dimension 2). Suppose also the spheres each have radius $\rho = 1/2$. Then the volume of one sphere is $\pi \cdot (1/2)^2 = \pi/4$. Previously, we found $\det \Lambda$ to be $3/4$. Thus, the density is $\Delta = \frac{\pi/4}{\sqrt{3/4}} = \frac{\pi}{\sqrt{12}} \approx 0.907$.

Formula. The volume of an n -dimensional sphere of radius ρ is $V_n \rho^n$, where

$$V_n = \frac{\pi^{n/2}}{(n/2)!} = \frac{2^n \cdot \pi^{(n-1)/2} \cdot ((n-1)/2)!}{n!}$$

Formula. The surface area of a sphere of radius ρ is $n V_n \rho^{n-1}$.

A *periodic* packing is one that is obtained by placing a fixed configuration of s spheres in each fundamental region of a lattice Λ .

Proposition. The density of a periodic packing is given by

$$\Delta = \frac{s \cdot (\text{volume of one sphere})}{\sqrt{\det \Lambda}}$$

The densest known sphere packing in 4 dimensions is the checkerboard lattice D_4 , which consists of all spheres that have centers at (u_1, u_2, u_3, u_4) where $u_i \in \mathbb{Z}$ and $\sum_{i=1}^4 u_i \equiv 0 \pmod{2}$.

Formula. The density of a lattice packing Λ is given by $\Delta = \frac{V_n \rho^n}{\sqrt{\det \Lambda}}$.

We say that if a lattice can be obtained from another by (possibly) a rotation, reflection, or change of scale, then we say that the lattices are equivalent and will $\cong$ to denote equivalent lattices. Look now at how this is related to the generator matrices. Let M_1, M_2 be the generator matrices of lattices Λ_1, Λ_2 , respectively. Then we say that Λ_1 is equivalent to Λ_2 (write $\Lambda_1 \cong \Lambda_2$) iff $M_2 = cUM_1B$, where c is a nonzero constant, U is a matrix with integer entries with $\det U = \pm 1$, and B is a real orthogonal matrix (with $BB^t = I$, necessarily). Informally, it could be viewed that c represents the scaling factor, U represents reflection due its determinant being restricted to the set $\{-1, 1\}$ and its entries being in $\mathbb{Z}$, and B represents rotation. If U has determinant ± 1 and $c = 1$, then M_1, M_2 are *congruent* lattices (they are *directly congruent* if $\det U = 1$).

Proposition. Let Λ_1, Λ_2 be equivalent lattices with generator matrices M_1, M_2 , respectively. Then Λ_1 and Λ_2 have the same density.

Proof. Assume the lattices live in dimension n . It suffices to show $\frac{V_n \rho_1^n}{\sqrt{\det \Lambda_1}} \stackrel{(*)}{=} \frac{V_n \rho_2^n}{\sqrt{\det \Lambda_2}}$. We may assume $V_n \neq 0$ and that M_1, M_2 are square. Suppose $i \in \{1, 2\}$. By definition of the generator matrix, the rows of M_i are linearly independent, so M_i is invertible. A calculation reveals $\det \Lambda_i = (\det M_i)^2$, so $\sqrt{\det \Lambda_i} = \det M_i$. Since $\Lambda_1 \cong \Lambda_2$, thus $M_2 = cUM_1B$, with c, U , and B suitably defined. Thus, $\det \Lambda_2 = (\det M_2)^2 = (\det(cUM_1B))^2 = (\det(cU) \det(M_1) \det(B))^2 = c^{2n} (\det M_1)^2$. So, $\sqrt{\det \Lambda_2} = c^n (\det M_1)$. Thus, $(*) \iff \frac{\rho_1^n}{\det M_1} = \frac{\rho_2^n}{c^n \cdot \det M_1} \iff c^n = \left(\frac{\rho_2}{\rho_1}\right)^n \iff c = \frac{\rho_2}{\rho_1}$, which is precisely the definition of the change of scale from Λ_1 to Λ_2 . $\square$

Any n -dimensional lattice L_n has a *dual* lattice L_n^* , given by $L_n^* = \{x \in \mathbb{R}^n \mid x \cdot u = \mathbb{Z} \forall u \in L_n\}$.

We have another way to measure density, as given by the *center density* $\delta := \frac{\Delta}{V_n}$. Intuitively, if the spheres in a lattice Λ each have radius 1, then δ represents the number of centers present in a given unit volume. From previously, we had $\Delta = \frac{V_n \rho^n}{\sqrt{\det \Lambda}}$, so the center density is $\delta = \frac{\rho^n}{\sqrt{\det \Lambda}}$.

Theorem (Minkowski). There exist lattices with density satisfying $\Delta \geq \frac{\zeta(n)}{2^{n-1}}$, where $\zeta(n)$ is the Riemann-zeta function.

We also introduce *Hermite's constant* γ_n , which is given by $\gamma_n := 4 \cdot \delta_n^{2/n}$, where δ_n is the center density of the densest lattice packing in $\mathbb{R}^n$.

Now we observe the following problems. The thirteen spheres problem asks: is it possible to arrange thirteen spheres (of identical size, in dimension 3) around another so that the thirteen spheres touch it simultaneously? The answer is 12, so any arrangement of spheres in such a way described in the problem will only hold if there are $n \leq 12$ spheres surrounding a center sphere. This motivates us to define the *kissing number* (denoted by τ) of a sphere packing (in any dimension), which is the number of spheres that touch one sphere. The kissing number problem has

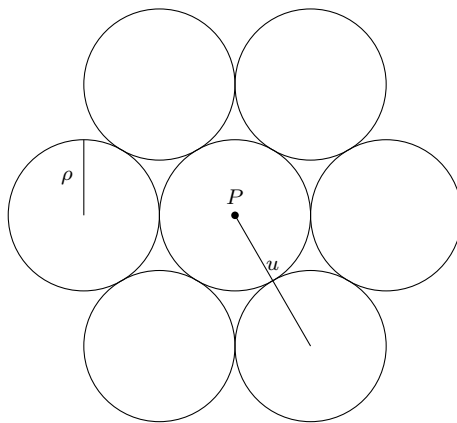
been solved in nontrivial dimension 8 and 24, due to the fact that the corresponding arrangements of the spheres in these dimensions are unique. (The arrangement in the 8-dimensional case is found in the E_8 lattice.) For notation, we write τ_n to denote the highest kissing number in dimension n .

We now discuss an equivalent of the kissing number problem. Consider the following question: how many points can be placed on the surface of a sphere in $\mathbb{R}^n$ so that the angular separation between any two points is at least 60° ? Denote the surface of a sphere Ω_n , where $\Omega_n = \{(x_1, \dots, x_n) \mid \sum x_i^2 = 1\}$. We define a *spherical code* to be a finite subset $X \subset \Omega_n$ and say that X has *minimal angle* ϕ if ϕ is the largest number for which $x \cdot y \leq \cos \phi$ for any $x, y \in X$ (with $x \neq y$). The kissing number problem generalizes to asking what the maximal number $A(n, \phi)$ of points in a spherical code is in Ω_n of minimal angle ϕ .

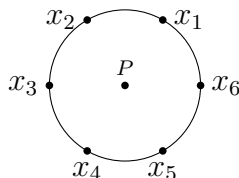
The construction of spherical codes from sphere packings will be given as follows. Let Λ to be a sphere packing in $\mathbb{R}^n$ that contains N spheres, P be the origin, and suppose the spheres are at a distance u from P . It thus follows that when “dividing the centers” by u (rescaling) gives an n -dimensional spherical code of size N in Ω_n .

Proposition. Consider the construction of a spherical code, as discussed above. Then the centers of the spheres are at least 2ρ units apart, where ρ is the radius of one sphere. Also, the minimal angle is at least $2 \arcsin(\rho/u)$.

We now consider an example. Let Λ be the following sphere packing in $\mathbb{R}^2$.



The resulting spherical code is the set $X = \{x_1, \dots, x_6\}$ pictured as follows:



We have that each x_i is a “kissing point”. We now calculate the minimal angle ϕ of X . Fix $x_i, x_j \in X$ with $i \neq j$. Letting $\rho = 1$ (thus $u = 2$) gives $x_i \cdot x_j = \|x_i\| \|x_j\| \cos \theta$, where θ is the angular separation of x_i and x_j . Since $i \neq j$, thus $\theta \in \{\pi/3, 2\pi/3, \pi\}$

in modulo π . If $\theta = \pi/3$, then $x_i \cdot x_j = 1/2$; if $\theta = 2\pi/3$, then $x_i \cdot x_j = -1/2$; if $\theta = \pi$, then $x_i \cdot x_j = -1$. Thus all $x_i \cdot x_j$ lie in the set $\{-1, -1/2, 1/2\}$. Thus, we conclude that the minimal angle is $\pi/3$, and indeed, $\pi/3 \geq 2 \arcsin(1/2)$.

2 Geometry & Theta Series

We define the *thickness of a covering* Θ to be the average number of spheres that contain a point of the space. This is equal to $\frac{\text{volume of one sphere}}{(\det \Lambda)^{1/2}} = \frac{V_n \cdot R^n}{(\det \Lambda)^{1/2}}$, where we assume an arrangement of spheres of radius R that covers $\mathbb{R}^n$. We also define the *normalized thickness* to be $\theta := \frac{\Theta}{V_n}$.

The covering problem asks the following question: what is the thinnest covering of an n -dimensional space by spheres? Take now a discrete collection of points $\mathcal{P} = \{P_1, P_2, \dots\}$ in $\mathbb{R}^n$. Define the *covering radius of $\mathcal{P}$* to be the least upper bound for the distance from any point of $\mathbb{R}^n$ to the closest point P_i , i.e.

$$R := \sup_{x \in \mathbb{R}^n} \inf_{P \in \mathcal{P}} \text{dist}(x, P)$$

where R is the symbol to denote the covering radius. In other words, R is the maximum possible distance a point $x \in \mathbb{R}^n$ can be from the lattice. If no such finite R exists, then we assign R to the value ∞ . For example, consider the set of 4 points $\mathcal{P} = (\pm 1, \pm 1)$ in $\mathbb{R}^2$. Then there exists a point in $\mathbb{R}^2$ of arbitrary distance to the lattice (we define distance from a point in $\mathbb{R}^n$ to a lattice Λ by the infimum of the distances from the point to each point in the lattice). Thus, $R = \infty$. If instead we chose the lattice $\mathbb{Z}^2$ in $\mathbb{R}^2$, then $R = \sqrt{2}$.

Now, for each point $P_i \in \mathcal{P}$, define its *Voronoi cell* $V(P_i)$ to be those points of $\mathbb{R}^n$ that are at least as close to P_i as to any other P_j , i.e.

$$V(P_i) = \{x \in \mathbb{R}^n \mid \text{dist}(x, P_i) \leq \text{dist}(x, P_j)\}$$

In other words, the Voronoi cell of a point $P \in \mathcal{P}$ consists of all those points $x \in \mathbb{R}^n$ whose closest point in $\mathcal{P}$ is P . Thus, the volume of the fundamental region of a lattice Λ is precisely the volume of a Voronoi cell, $\sqrt{\det \Lambda}$, where all Voronoi cells in the lattice are congruent. The vertices of the Voronoi cells include the points of $\mathbb{R}^n$ whose distance from $\mathcal{P}$ is a local maximum, and we call these points *holes* in $\mathcal{P}$. If there is a point whose distance from $\mathcal{P}$ is an absolute maximum, it is called a *deep hole*. Holes that are not deep are called *shallow holes*.

For a lattice Λ , with Voronoi cells congruent to some polytope V , the packing radius ρ is the *inradius* of V (the radius of the largest inscribed sphere), while the covering radius R is the *circumradius* of V (the radius of the smallest circumscribed sphere).

The *minimalized second moment* of $V(P_i)$ is defined to be

$$G = G(V(P_i)) = \frac{1}{n} \text{Vol}(V(P_i))^{-1-\frac{2}{n}} \int_{V(P_i)} \|x - P_i\|^2 dx$$

Consider the analogy as follows. Let the P_i be schools and the Voronoi cells the school districts. Then, G is the average squared distance that a child who lives in $V(P_i)$ has to walk to school. The points $P_{i1}, \dots, P_{ir}$ such that the hyperplane (a generalization of a 2D plane in 3D-space) between P_i and P_{ij} contains a face of $V(P_i)$ with positive area are called *Voronoi-relevant* for P_i . For a lattice Λ , the *Voronoi-relevant vectors* $v \in \Lambda$ are those that are relevant for the origin, i.e. that are needed to define $V(0)$. The *Delaunay cell* for each point that is the vertex of a Voronoi cell is the polytope which is the convex hull of the points of $\mathcal{P}$ closest to that point, where a convex hull of points is the smallest convex set that encloses all the points.

We now discuss quadratic forms. The *squared length* (or *norm*) of a vector is given by $N(x) = x \cdot x = (x, x) = \sum x_i^2$. The *minimal norm of a lattice* Λ is the minimal squared distance between distinct lattice vectors, given by $\min\{N(x) \mid x \in \Lambda, x \neq 0\}$. Take now Λ to be a lattice in $\mathbb{R}^n$, with basis vectors $v_1, \dots, v_n$. Take $x = (x_1, \dots, x_n)$ to be a vector in Λ . Thus, since x lies in the integral span of $v_1, \dots, v_n$, thus there exist $\xi_1, \dots, \xi_n \in \mathbb{Z}$ so that $x = \xi_1 v_1 + \dots + \xi_n v_n$. Thus each $x \in \Lambda$ can be characterized uniquely by an n -tuple of integers. The norm of x is given by $N(x) = N(\xi_1 v_1 + \dots + \xi_n v_n) = \sum_{i=1}^n \sum_{j=1}^n \xi_i \xi_j v_i \cdot v_j$, which is precisely $\xi M M^t \xi^t = \xi A \xi^t = f(\xi)$, where $\xi = (\xi_1, \dots, \xi_n)$. We call $f(\xi)$ a *quadratic form* associated with the lattice, since it is a function of the n integer variables. Examples of quadratic forms include $\xi_1^2 + \xi_1 \xi_2 + \xi_2^2$ and $\xi_1^2 + \dots + \xi_n^2$.

Previously, we introduced the concept of congruent lattices. Quadratic forms corresponding to congruent lattices are *integrally equivalent*, i.e. there exists a 1-1 correspondence between congruence classes of lattices and integral equivalence classes of quadratic forms. Consider now lattices Λ_1, Λ_2 with bases $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1/2 \\ \sqrt{3}/2 \end{pmatrix}$ and $\begin{pmatrix} \sqrt{3}/2 \\ 1/2 \end{pmatrix}, \begin{pmatrix} \sqrt{3} \\ 0 \end{pmatrix}$, respectively. Then the generator matrices are $\begin{pmatrix} 1 & 0 \\ -1/2 & \sqrt{3}/2 \end{pmatrix}$ and $\begin{pmatrix} \sqrt{3} & 0 \\ \sqrt{3}/2 & 1/2 \end{pmatrix}$, respectively. Thus the Gram matrices for Λ_1 and Λ_2 are $A_1 = \begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix}$ and $A_2 = \begin{pmatrix} 3 & 3/2 \\ 3/2 & 1 \end{pmatrix}$, respectively. Now we find the associated quadratic forms of the lattices. For Λ_1 , we calculate $\xi A_1 \xi^t$, where $\xi = \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix}$, thus giving $\xi A_1 \xi^t = \xi_1^2 - \xi_1 \xi_2 + \xi_2^2$. For Λ_2 , a similar calculation reveals that the quadratic form is $3\xi_1^2 + 3\xi_1 \xi_2 + \xi_2^2$. Through number-theoretic methods it can be verified that the quadratic forms are equivalent (by finding an element in the modular group that takes one form to the other) and thus the lattices that these quadratic forms represent are the same, up to a scaling, rotation, or reflection. Quadratic forms are useful in some cases in lattice theory, but we prefer to conduct further examination on the geometric aspects.

We now discuss theta series and connections with number theory. We may ask the following question: what is the number of ways we may write an integer m as a sum of 4 squares? We may answer this question with lattices. For a lattice Λ , let

N_m be the number of vectors $x \in \Lambda$ of norm m , i.e. with $x \cdot x = m$. Equivalently, N_m is the number of integral vectors ξ so that $\xi A \xi^t = m$. Notice that $x_1^2 + \dots + x_4^2$ is the quadratic form associated with the lattice $\mathbb{Z}^4$, as can be verified. Thus we see that the number of ways of writing m as a sum of 4 squares is equal to the number of vectors of norm m in $\mathbb{Z}^4$. We now introduce the *theta series* of a lattice (which can be used to calculate these numbers), defined by

$$\Theta_\Lambda(z) = \sum_{x \in \Lambda} q^{x \cdot x} = \sum_{m=0}^{\infty} N_m q^m$$

where $q = e^{\pi i z}$. The *average theta series* of a periodic packing $\mathcal{P}$ (consisting of s translates $u_j + \Lambda, j = 1, \dots, s$) is defined to be

$$\begin{aligned} \Theta_{\mathcal{P}}(z) &= \frac{1}{s} \sum_{j=1}^s \sum_{k=1}^s \sum_{x \in \Lambda} q^{N(x+u_j-u_k)} \\ &= \Theta_\Lambda(z) + \frac{2}{s} \sum_{j < k} \sum_{x \in \Lambda} q^{N(x+u_j-u_k)} \end{aligned}$$

If $\mathcal{P}$ is *distance invariant* (meaning it has the property that the number of its points at any distance d from a point $x \in \mathcal{P}$ is independent of x) then $\Theta_{\mathcal{P}}(z)$ reduces to $\Theta_{\mathcal{P}}(z) = \sum_{j=1}^s \sum_{x \in \Lambda} q^{N(x+u_j-u_1)}$.

Proposition. The norm function $N(x)$ has the following property:

$$\begin{aligned} N((x_1, \dots, x_n)) &= x_1^2 + \dots + x_n^2 \\ &= N(x_1) + \dots + N(x_n) \end{aligned}$$

The theta series of a lattice tells us the packing radius ρ , the kissing number τ , and the density Δ , since $\Theta_\Lambda(z) = 1 + \tau q^{4\rho^2} + \dots$ and $\Delta = (\lim_{n \rightarrow \infty} (\rho/r)^n) \cdot \sum_{m \leq r^2} N_m$. We may obtain the theta series of the dual lattice Λ^* by

$$\Theta_{\Lambda^*}(z) = \sqrt{\det \Lambda} (i/z)^{n/2} \Theta_\Lambda(-1/z)$$

So far we have determined a theta series for a given lattice. However, in general, it is not possible to construct a unique lattice from a given theta series. To see this we may consider the lattice D_{16}^+ and the direct sum $E_8 \oplus E_8$. (Note that D_n is the lattice defined by the collection of points in $\mathbb{Z}^n$ whose coordinates sum to an even number, and $D_n^+ := D_n \cup (1/2, \dots, 1/2) + D_n$. Also, D_n^+ is a lattice iff n is even.) D_4^+ is congruent to $\mathbb{Z}^4$, as can be verified. Thus, we have the equation

$$\frac{1}{2}(\theta_2(z)^4 + \theta_3(z)^4 + \theta_4(z)^4) = \theta_3(z)^4$$

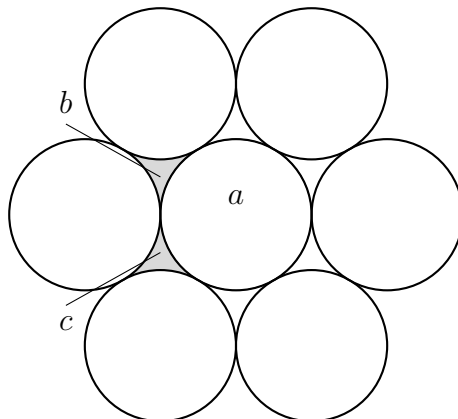
so $\theta_3(z)^4 = \theta_2(z)^4 + \theta_4(z)^4$, where $\theta_2, \theta_3, \theta_4$ are theta functions. The equality of the theta series of D_{16}^+ and $E_8 \oplus E_8$ follows from showing the following equation holds:

$$\frac{1}{2}(\theta_2(z)^{16} + \theta_3(z)^{16} + \theta_4(z)^{16}) = \frac{1}{4}(\theta_2(z)^8 + \theta_3(z)^8 + \theta_4(z)^8)$$

Now, we discuss integral lattices and quadratic forms. We call a lattice or quadratic form *integral* if the inner product of any two lattice vectors is an integer, i.e. the Gram matrix has integer entries.

Proposition. A lattice Λ is integral iff $\Lambda \subseteq \Lambda^*$.

The *dual quotient group* Λ^*/Λ has order $\det \Lambda$. As an example, consider the lattice A_2 :



The elements A_2^*/A_2 are a, b, c , where a represents a circle and b, c represent “downward” and “upward” gaps, respectively.

Proposition. If Λ is an integral lattice, then $\Lambda \subseteq \Lambda^* \subseteq \frac{\Lambda}{\det \Lambda}$.

An integral lattice Λ with $|\det \Lambda| = 1$ (or equivalently, with $\Lambda = \Lambda^*$) is called *unimodular* or *self-dual*. If $x \cdot x$ is even for all $x \in \Lambda$, then Λ is *even*; otherwise, we say that Λ is *odd*. Taking the lattice $\mathbb{Z}^2$, we see that $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1$, so $\mathbb{Z}^2$ is odd. An example of an even unimodular lattice is E_8 , whose construction we will observe later.

Theorem. If a unimodular lattice has the property that the norm of every lattice vector is a multiple of some positive integer c , then $c \in \{1, 2\}$.

The *automorphism group* of a lattice Λ is defined to be the set of distance-preserving transformations (isometries) of the space that fix the origin and take the lattice to itself.

Smith-Minkowski-Siegel Mass Formula. Let $|\text{Aut } \Lambda|$ be the order of the automorphism group of a lattice Λ . Then there is a constant a_0 for which $a_0 = \sum_{\Lambda} \frac{1}{|\text{Aut } \Lambda|}$, where the sum is over all nonequivalent lattices from a given genus (an example of a genus is all n -dimensional odd unimodular lattices).

We can use this formula to check the number of unimodular lattices of dimension n containing no vectors of norm 1, and other such information.

A *modular form* of weight k is a holomorphic function (on the upper-half plane)

$f : \mathbb{H} \rightarrow \mathbb{C}$ with the following properties:

1. If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$ so that $A : \tau \mapsto \frac{a\tau+b}{c\tau+d}$, then $f(A(\tau)) = (c\tau + d)^k f(\tau)$.
2. For any $\gamma \in SL_2(\mathbb{Z})$, the function $(cz + d)^{-k} f(\gamma(z))$ is bounded for $\text{Im} z \rightarrow \infty$.

A key point is that the theta series of an integral lattice is a modular form.

3 Theta Series Coefficients

In this section, we outline a procedure to compute the terms of the theta series of a given lattice. Recall from earlier that for a given lattice Λ , the m th term in its theta series is the number of vectors in Λ with norm m , where the norm we refer to is the “squared Euclidean length” norm.

First, assume a lattice Λ has dimension d , and choose a basis $b_1, \dots, b_d$ of the lattice. Then write a matrix B with columns the elements b_i . We now apply the Lenstra-Lenstra-Lovász (LLL) lattice basis reduction algorithm. The purpose of conducting LLL-reduction is to modify the existing basis $b_1, \dots, b_d$ so that the basis vectors are of shorter length and together form a basis close to orthogonal. Next, we calculate the Cholesky factorization of B so that further calculations may be executed more efficiently. Then, we use algorithm 2.12 by Fincke & Post (paper link) to calculate the solution the number of lattice vectors of norm m within a given search region.

4 Lattice Examples

In this section, we will discuss some important lattices. First, we discuss the theory of gluing. The idea behind gluing theory is that we want to describe the general n -dimensional integral lattice L that has a sublattice which is a direct sum $L_1 \oplus \dots \oplus L_k$ of given integral lattices $L_1, \dots, L_k$ of total dimension n . If $y \in L$, then we can write $y = y_1 + \dots + y_k$, where y_i is in the subspace spanned by L_i , although not necessarily in L_i . We want to learn about the possibilities for the y_i . If $l_i \in L_i$, then $y_i \cdot l_i \in \mathbb{Z}$ has the same value as $y \cdot l_i$, since y is an element of the direct sum. Thus we may conclude that $y_i \in L^*$, where L^* is the dual lattice of L . Since any y_i can be altered by adding a vector of L_i , we may suppose that y_i is one of a standard system of representatives for the cosets of L_i in L_i^* . These representatives are called *glue vectors* for L_i . For ease, we choose the glue vectors to be of minimal length in their cosets. The quotient group L_i^*/L_i is called the *dual group* or *glue group* for L_i . We say that a lattice L is formed by *gluing the components* $L_1, \dots, L_k$ by the glue vectors $y_1, \dots, y_k$. In the case that only one of the glue vectors y_i is nonzero, we say that its corresponding L_i has *self-glue* and that y is a *self-glue* vector.

Theorem. If a unimodular lattice L is formed by gluing together two lattices L_1 and L_2 in such a way that there is no self-glue, i.e.

$$L_1 = (L_1 \otimes \mathbb{R}) \cap L,$$

$$L_2 = (L_2 \otimes \mathbb{R}) \cap L$$

then the dual quotients L_1^*/L_1 and L_2^*/L_2 are isomorphic groups.

We now discuss more about theta functions. Consider a glue group L^*/L , of order $\det L$, with glue vectors indicated by square brackets $[\cdot]$. Let $G(L)$ be the automorphism group of L , G_{01} be the group of automorphisms in $G(L)$ that give the trivial permutation, and $G_0(L)$ be the normal subgroup of G_{01} consisting of those automorphisms which, for every i , send each glue vector y_i into a vector to a vector in the same coset $y_i + L_i$, i.e. which fix the glue vectors modulo the components. Let $G_1(L)$ denote the group that is isomorphic to a permutation group acting on the glue vectors of each component. If $g_0 = |G_0(L)|$, $g_1 = |G_1(L)|$, then $g := |G(L)| = g_0 \cdot g_1$.

Proposition. Let L be a lattice and $\Theta_L(z)$ be its theta series in powers of q , where $q = e^{i\pi z}$. Thne the coefficients in the expansion of $\Theta_L(z)$ give the numbers of vectors in the successive shells of L around the origin, i.e. the coefficients in the expansion of $\Theta_L(z)$ give the number of vectors in the lattice L of a given norm.

We now introduce the Jacobi theta function:

$$\theta_3(\xi | z) = \sum_{m=-\infty}^{\infty} e^{2mi\xi + \pi izm^2}, \text{Im}z > 0$$

Now, we relate lattices that are rescaled versions of each other. Let Λ, Λ' be lattices so that $\Lambda' = c\Lambda = \{cx \mid x \in \Lambda\}$ for some constant $c \in \mathbb{R}$. Then we have that $M' = cM$, $A' = c^2A$, $\det \Lambda' = c^{2n} \det \Lambda$, $(\text{minimal norm})' = c^2 \cdot (\text{minimal norm})$, $\rho' = c\rho$, and $R' = cR$, where R is the covering radius. The kissing numbers τ , densities Δ, δ , and thickness are unchanged. Additionally, $(\Lambda')^* = (1/c)\Lambda^*$ and $\Theta_{\Lambda'}(z) = \Theta_{\Lambda}(c^2z)$.

We discuss now the lattice $\mathbb{Z}^n = \{(x_1, \dots, x_n) \mid x_i \in \mathbb{Z} \forall i\}$. This is called the n -dimensional *cubic* or *integer lattice*. The theta series of $\mathbb{Z}^n$ is $\theta_3(z)^n$. We have $\Theta_{\mathbb{Z}^n}(z) = \sum_{m=0}^{\infty} N_m q^m$. Taking $n = 2$ gives $\Theta_{\mathbb{Z}^2}(z) = 1q^0 + 4q^1 + 4q^2 + 0q^3 + 4q^4 + 8q^5 + \dots = \theta_3(z)^2$. The theta series of the translate lattice $\mathbb{Z}^n + (0^a(\frac{1}{2})^{n-a})$ is $\theta_2(z)^{n-a}\theta_3(z)^a$. In the $n = 2$ case, we notice that the m th coefficient in the theta series is precisely the number of ways of writing m has a sum of two squares. It follows that the theta series is $\sum_{m=0}^{\infty} r_n(m)q^m$, where $r_n(m)$ is the number of ways of writing m has a sum of n squares.

We define now the lattice A_n to be the set $\{(x_0, \dots, x_n) \in \mathbb{Z}^{n+1} \mid x_0 + \dots + x_n = 0\}$. We have a generator matrix M for A_n , given below:

$$M = \begin{pmatrix} -1 & 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & -1 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & -1 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & -1 & 1 \end{pmatrix}$$

It can be verified that $\det A_n = n + 1$. Taking $n = 2$ gives that a generator matrix is $M = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$. It follows that the A_n lattice lies in the hyperplane $\sum x_i = 0$

in $\mathbb{R}^{n+1}$. Clearly the minimal vectors of A_n are the vectors that are permutations of $(-1, 1, 0, \dots, 0)$, since we require these vectors to be of minimal norm and that each component of a vector must lie in $\mathbb{Z}$.

The lattice A_2 is equivalent to the hexagonal lattice. One way to visualize this is by taking the hexagonal lattice in $\mathbb{R}^2$, viewing it in $\mathbb{R}^3$, and then performing scaling/rotating operations. Consider now a generator matrix $M = \begin{pmatrix} 1 & 0 \\ -1/2 & \sqrt{3}/2 \end{pmatrix}$ for the hexagonal lattice. The corresponding Gram matrix is $A = \begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix}$, and so the associated quadratic form is $\xi_1^2 - \xi_1\xi_2 + \xi_2^2$. Thus, the theta series for the hexagonal lattice is $\sum_{x,y \in \mathbb{Z}} q^{x^2 - xy + y^2}$.

A non-example of a lattice is the packing $D_3^+ = D_3 \cup (1/2, 1/2, 1/2) + D_3$, where $D_3 = \{(x, y, z) \in \mathbb{Z}^3 \mid x + y + z \equiv 0 \pmod{2}\}$. Points in D_3^+ include $(0, 0, 0)$, $(2, 4, 2)$, $(2.5, 4.5, 2.5)$, $(10.5, 10.5, 10.5)$, etc. To see why D_3^+ is not a lattice, take the points $u = (10, 8, 6)$ and $v = (2.5, 2.5, 2.5)$. Their difference is $(7.5, 5.5, 3.5) = (1/2, 1/2, 1/2) + (7, 5, 3)$, where $(7, 5, 3) \notin D_3$.

We now introduce the lattices E_8 , E_7 , and E_6 . The E_8 lattice is defined to be D_8^+ (in the “even” coordinate system). Another convenient way of writing E_8 is $\{(x_1, \dots, x_8) \mid x_i \in \mathbb{Z}\forall i \text{ or } x_i \in \mathbb{Z} + 1/2\forall i, \sum x_i \equiv 2x_8 \pmod{2}\}$. (The latter is the E_8 lattice viewed in the “odd” coordinate system.) A generator matrix for E_8 (in the even coordinate system) is given below:

$$M = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 \end{pmatrix}$$

The theta series is given by $\Theta_{E_8}(z) = \sum_{m=0}^{\infty} N_m q^m = 1 + 240q^2 + 2160q^4 + \dots$, where $N_m = 240\sigma_3(m/2)$, where $\sigma_r(m) = \sum_{d|m} d^r$.

The E_7 lattice is defined to be the set $\{x \in E_8 \mid \exists v \in E_8 : x \cdot v = 0\}$, where we require v to be a vector of minimal norm in E_8 . We define the lattice E_6 to be the set $\{x \in E_8 \mid x \cdot v = 0\forall v \in V\}$, where V is an A_2 -sublattice in E_8 .

Now, we introduce the 24-dimensional Leech lattice Λ_{24} . We first make the following definitions. The *extended binary Golay code* $\mathcal{C}_{24}$ is a vector subspace W of the space $V = \mathbb{F}_2^{24}$ of 24-bit words so that any two distinct elements of W differ in at least 8 coordinates. Elements of W are called *codewords*. We now give a construction of the Leech lattice. Λ_{24} is generated by all vectors of the form $\frac{1}{\sqrt{8}}(\mp 3, \pm 1^{23})$, where the ∓ 3 may be in any position, and the upper signs are taken on a set of coordinates where a codeword of the binary Golay code $\mathcal{C}_{24}$ is 1. We now give another construction of the Leech lattice. Λ_{24} consists of the vectors $a(0 + 2c + 4x)$, $a(1 + 2c + 4y)$, where

$a = 1/\sqrt{8}$, $0 = (0^{24})$, $1 = (1^{24})$, $c \in \mathcal{C}_{24}$ (regarding the components of c as “real” 0’s and 1’s rather than elements of $\mathbb{F}_2$), and $x, y \in \mathbb{Z}^{24}$ satisfy $\sum x_i \equiv 0 \pmod{2}$ and $\sum y_i \equiv 1 \pmod{2}$. Vectors of the form $a(0 + 2c + 4x)$ are called the *even* vectors and vectors of the form $a(1 + 2c + 4y)$ are called the *odd* vectors. The theta series for Λ_{24} is

$$\begin{aligned}\Theta_{\Lambda_{24}}(z) &= \sum_{m=0}^{\infty} N(m)q^m \\ &= 1 + 196560q^4 + 16773120q^6 + \dots\end{aligned}$$

where $N(m) = \frac{65520}{691}(\sigma_{11}(m/2) - \tau(m/2))$, where $\tau(k)$ is a Ramanujan number. (Ramanujan numbers are the smallest positive integers that are sums of 2 cubes of positive integers in n ways.)

5 Quadratic Forms

Put $x = (x_1 \ x_2)$ and $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$. Then, $f(x) = xAx^t = ax_1^2 + 2bx_1x_2 + cx_2^2$ is called the *binary quadratic form* with matrix A . The associated *bilinear form* is $f(x, y) = xAy^t = ax_1y_1 + bx_1y_2 + bx_2y_1 + cx_2y_2$, where x and y are vectors with 2 components each. We may generalize these definitions to obtain n -ary quadratic forms and n -ary symmetric bilinear forms. Consider the binary quadratic form $ax_1^2 + 2bx_1x_2 + cx_2^2$. This binary form is *integral as a quadratic form* if $a, 2b, c \in \mathbb{Z}$ and *integral as a symmetric bilinear form* if $a, b, c \in \mathbb{Z}$. The term *classically integral* is equivalent to the latter definition made in the previous sentence. In the study of lattices, we say that f is an *integral form* iff its matrix entries are integers.

We discuss now integral equivalence. Let V be a $\mathbb{Q}$ -vector space with inner product $(\cdot, \cdot)$ and say the norm of $x \in V$ is (x, x) . Suppose $\dim V = 2$ (with $V = \text{span}(e_1, e_2)$) and $a = (e_1, e_1)$, $b = (e_1, e_2)$, and $c = (e_2, e_2)$. Then the norm of $x = x_1e_1 + x_2e_2$ is $(x, x) = ax_1^2 + 2bx_1x_2 + cx_2^2 = f(x)$, and the inner product $(x, y = y_1e_1 + y_2e_2)$ is equal to $f(x, y)$.

Fix now a basis e_1, e_2 of V . The collection $\{x_1e_1 + x_2e_2 \mid x_1, x_2 \in \mathbb{Z}\}$ forms a *lattice* in V and e_1, e_2 is an *integral basis* for this lattice. The other integral bases for this lattice have the form $\alpha e_1 + \beta e_2, \gamma e_1 + \delta e_2$, where $\alpha, \beta, \gamma, \delta \in \mathbb{Z}$ with $\alpha\delta - \beta\gamma = \pm 1$. We call the binary forms f, g *integrally equivalent*, or say they are in the same *class*, and write $f \sim g$, if there exists a matrix $M = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ with $\alpha, \beta, \gamma, \delta \in \mathbb{Z}$ and $\det M = \pm 1$ for which $B = MAM^t$, where f, g have matrices A, B , respectively. As seen in a previous section, f, g refer to the same lattice with different integral bases. The equivalence $f \sim g$ is *proper* if $\det M = 1$ and *improper* if $\det M = -1$. These definitions generalize seamlessly in dimension n .

Proposition. If f, g are integrally equivalent, then $\det A = \det B$.

Proof. By definition, there exists a matrix $M = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ with $\alpha, \beta, \gamma, \delta \in \mathbb{Z}$ and $\det M = \pm 1$ so that $B = MAM^t$. Then $\det B \stackrel{(*)}{=} \det M \cdot \det A \cdot \det M^t = \det A$, as $\det M = \det M^t$, and $(*)$ holds since elements of $\{-1, 1\}$ square to 1. $\square$

If f is a form with matrix A , then we can transform f into an equivalent form with matrix B by performing elementary row operations on A and then performing the exactly corresponding column operations. (Row operations consist of either multiplying a row by a unit or adding multiples of one row to another.)

We now move on to the classification of binary quadratic forms.

Theorem. Let $\begin{pmatrix} a_0 & b_0 \\ b_0 & a_1 \end{pmatrix}$ (with abbreviation $a_0^{b_0} a_1$) be an integral binary quadratic form of determinant $d = a_0 a_1 - b_0^2$ and let $-d$ not be a square. Define a sequence

$$\begin{pmatrix} a_0 & b_0 \\ b_0 & a_1 \end{pmatrix}, \dots, \begin{pmatrix} a_i & b_i \\ b_i & a_{i+1} \end{pmatrix}, \dots$$

of binary forms of discriminant d by the following rules:

1. a_i and b_i determine a_{i+1} by $\frac{b_i^2 + d}{a_i}$.
2. b_i and a_{i+1} determine b_{i+1} as the largest solution of $b_i + b_{i+1} \equiv 0 \pmod{a_{i+1}}$ for which $b_{i+1}^2 + d < 0$ if such solutions exist, and otherwise as the smallest solution of the congruence in absolute value, taking b_{i+1} positive in case of a tie.

Then the sequence of forms derived in this way from the given form is ultimately periodic, the forms in the period being called a *cycle of reduced forms*. Also, two binary forms are properly equivalent iff they lead to the same cycle of reduced forms.

Consider the following example. Let $f = x^2 - 67y^2$, with matrix $\begin{pmatrix} a_0 & b_0 \\ b_0 & a_1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -67 \end{pmatrix}$. Applying the above procedure gives the following. $a_1 = -67$, so $b_0 + b_1 \equiv 0 \pmod{a_1}$ becomes $b_1 \equiv 0 \pmod{67}$, thus giving $b_1 = 0$, as $d = -67$ and $b_1^2 + d < 0$ is satisfied with $b_1 = 0$. Then $a_2 = \frac{b_1^2 + d}{a_1} = \frac{-67}{-67} = 1$, so the second term of the sequence is $\begin{pmatrix} -67 & 0 \\ 0 & 1 \end{pmatrix}$. Then $a_2 = 1$, so we now solve $b_1 + b_2 \equiv 0 \pmod{1}$, giving $b_2 \equiv 0 \pmod{1}$, for which $b_2^2 - 67 < 0$. Notice any value in the complete residue system modulo a_2 can be satisfied by b_2 , so the restriction $b_2^2 - 67 < 0$ resolves b_2 to 8. Thus, $a_3 = \frac{b_2^2 - 67}{a_2} = \frac{-3}{1} = -3$, so the third term of the sequence is $\begin{pmatrix} 1 & 8 \\ 8 & -3 \end{pmatrix}$. Through further calculation, it can be verified that, eventually, the obtained sequence is 10-periodic.

In number theory, a positive definite form $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ is reduced iff $-a < 2b \leq a \leq c$,

with $b \geq 0$ if $a = c$. Geometrically, we interpret this as saying that a is the absolutely smallest value taken (at a vector e_1 say) by the form and c is the absolutely smallest value that can be taken by the form at a vector e_2 independent of e_1 .

We consider now the compositions of binary forms. Suppose $-d$ is square-free with $-d \not\equiv 1 \pmod{4}$. Observe $\mathbb{Z}[\sqrt{-d}] \subseteq \mathbb{Q}[\sqrt{-d}]$. The *properly primitive* quadratic forms are those $f(x, y) = ax^2 + 2bxy + cy^2$ for which $\gcd(a, 2b, c) = 1$. Consider the properly primitive quadratic forms of determinant $ac - b^2 = d$. Let $I = \langle r + s\sqrt{-d}, t + u\sqrt{-d} \rangle$ be an ideal in $\mathbb{Z}[\sqrt{-d}]$. We define its *norm* to be $|ru - st|$. A *principal ideal* is generated by $r + s\sqrt{-d}$ and $(\sqrt{-d})(r + s\sqrt{-d})$ (over $\mathbb{Z}$), with norm $r^2 + s^2d$. A *principal normed ideal* is $\langle r + s\sqrt{-d} \rangle_{r^2 + s^2d}$. Multiplication of normed ideals is defined as $I_N \cdot J_M = (IJ)_{NM}$. Two (normed) ideals I, J are said to be in the same (normed) *ideal class* iff there exist principal (normed) ideals P, Q for which $IP = JQ$, and under composition, these ideal classes form a group, the (normed) *ideal class group*.

We now show that the proper (i.e. determinant 1) equivalence classes of properly primitive forms of determinant d correspond to classes of normed ideals. Suppose again $-d$ is square-free with $-d \not\equiv 1 \pmod{4}$. The normed ideal $\langle a, -b + \sqrt{-d} \rangle_a$ corresponds to f , where $\frac{x}{y} = \frac{-b + \sqrt{-d}}{a}$ is a solution of the equation $f(x, y) = ax^2 + 2bxy + cy^2 = 0$. We now define *composition*. Let f, g be quadratic forms. We compose f and g by multiplying their corresponding normed ideals, and by working modulo principal normed ideals, converting the product to a normed ideal of the shape $\langle a, -b + \sqrt{-d} \rangle_a$, and reduce the quadratic form $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ (with determinant d) by the procedure outlined in the last theorem.